







estimate that U_1
may also like M_3 .

Find similar users or items
Predict from neighbours

User-KNN • Item-KNN

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Compress users and items
Predict from learned vectors

Matrix factorization • SVD

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Same evidence: neighbours now, or learned structure first.





B

D



B

D

Prediction comes from observed neighbours,
without first learning a latent representation.

Model-based direction

Learn a compact
model once, then
reuse it for prediction.

readable patterns.

The model may discover axes such as “mainstream vs niche” or “light comedy vs dark drama” without human labels.



KNN

Memory-based
prediction



$k = 3$ neighbors

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$k = 3$ neighbors

The neighborhood provides local evidence, usually measured by cosine or Pearson similarity.

$$\min_{P,Q} \sum_{(u,i) \in \Omega} (r_{ui} - p_u^\top q_i)^2 + \lambda(\|p_u\|^2 + \|q_i\|^2)$$

KNN

Explainable,
weaker at scale

KNN

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MF / SVD

Compressed
latent factors

KNN

Explainable,
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systems

Hybrid: content
+ collaborative

**The system does not know exactly what we will like
it estimates from patterns shared across users, items
and context.**